DGMS Examination 2019 — Question 49 Detailed Solution

Junior Scientific Officer (Geology) | Volcanology & Igneous Geomorphology

QUESTION 49

Match List-I (Volcano type) with List-II (Characteristic):
List-I: A. Shield volcano | B. Cinder cone volcano | C. Stratovolcano | D. Lava dome volcano
List-II: 1. Also known as composite volcano | 2. Has shape of a flattened dome | 3. Formed exclusively of pyroclastic materials | 4. Formed by highly viscous lava

(a) A-4, B-1, C-3, D-2	(b) A-4, B-3, C-1, D-2
(c) A-2, B-3, C-1, D-4 CORRECT ANSWER	(d) A-2, B-1, C-3, D-4

1. Executive Summary & Answer Verification

The correct option is **(c) A-2, B-3, C-1, D-4**.

Volcanic edifices are morphologically and genetically classified on the basis of their eruptive mechanism, magma viscosity, and relative proportions of effusive lavas versus explosive pyroclastic ejecta:

- **A. Shield Volcano → 2 (Has shape of a flattened dome):** Formed by thousands of successive, highly fluid, low-viscosity basaltic lava flows that travel tens of kilometers from central/fissure vents, building an enormous, broad, convex-upward profile resembling a flattened warrior's shield (slopes typically 2°–10°, e.g., Mauna Loa).
- **B. Cinder Cone Volcano → 3 (Formed exclusively of pyroclastic materials):** Steep conical hills (slopes 30°–35° at the angle of repose) constructed exclusively or predominantly of vesicular, fragmental tephra (scoria, cinders, bombs) ejected during short-lived strombolian gas fountaining (e.g., Parícutin, Sunset Crater).
- **C. Stratovolcano → 1 (Also known as composite volcano):** Classic symmetrical conical peaks composed of alternating strata of viscous intermediate (andesitic/dacitic) lava flows and explosive pyroclastic fall/flow deposits; hence universally termed *composite volcanoes* (e.g., Mt. Fuji, Mt. Rainier, Mt. St. Helens).
- **D. Lava Dome Volcano → 4 (Formed by highly viscous lava):** Bulbous, steep-sided mounds or plugs formed by the sluggish extrusion of highly viscous, degassed, silica-rich (dacitic/rhyolitic) lava that cannot flow away from the vent (e.g., Lassen Peak, Novarupta dome, Mt. St. Helens dome).

pipe-like conduit through a subcircular surface hole (the vent). There may be a bowl- or funnel-shaped depression at the vent, called a **crater**. The rising magma follows weakened zones, fractures, or fracture intersections in the brittle, shallow crust. Magma solidifies in the vent and plugs it, strengthening the old weakness and forcing later eruptive phases to seek other conduits nearby. There may thus be more than one vent associated with a single volcano, particularly if we consider successive eruptions over a sufficient time span.

The 1980 eruption of Mt. St. Helens, for example, was dacitic and locally devastating. Whereas some volcanic centers may be rather monotonous, either in terms of composition, eruptive style, or both, others may show considerable variety. The underlying reasons for this difference are not clear.

Each eruptive phase will leave some solidified magma filling the central vent and conduit. Successive magmas

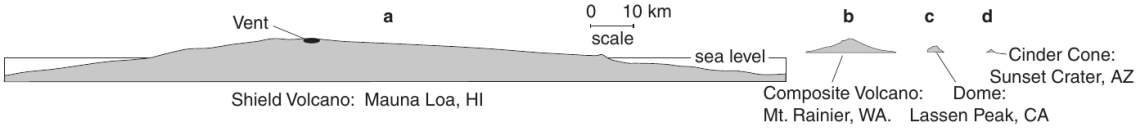


FIGURE 2 Volcanic landforms associated with a central vent (all at the same scale).

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Figure 49.1: Comparative profiles of the four primary volcanic edifices drawn to the same scale: (a) Shield Volcano (Mauna Loa, HI) showing its broad, flattened dome shape; (b) Composite Volcano / Stratovolcano (Mt. Rainier, WA); (c) Cinder Cone (Sunset Crater, AZ) built of loose tephra; and (d) Lava Dome (Lassen Peak, CA) built of viscous plug lava. (John D. Winter, Principles of Igneous and Metamorphic Petrology, 2nd ed., Chapter 4, Figure 2, p. 62).

2. Comprehensive Analysis of the Four Volcano Classes

2.1 Shield Volcanoes (Flattened Dome Morphology)

Shield volcanoes are the largest single volcanic landforms on Earth and in the Solar System (e.g., Olympus Mons on Mars). Key physical and petrological attributes include:

- **Magma Type:** Low-silica, high-temperature (1100°–1250°C) tholeiitic to alkali basalt.
- **Eruptive Style:** Non-explosive, effusive Hawaiian-type eruptions. Due to exceptionally low viscosity ($\sim 10\text{ to }100\text{ Pa}\cdot\text{s}$), lavas flow swiftly as pahoehoe and aa, traveling up to 50 km before solidifying.
- **Edifice Architecture:** Gentle slope profile ($<10^\circ$, averaging $2^\circ\text{--}5^\circ$) with a broad summit caldera. The ratio of basal diameter to height is massive ($>20:1$). For example, Mauna Loa rises ~9 km above the abyssal floor with a base diameter exceeding 100 km and a total basalt volume $>40,000\text{ km}^3$.

2.2 Cinder Cones (Pyroclastic / Scoria Cones)

Cinder cones are the simplest and most widespread volcanic landforms:

- **Constituent Material:** Formed *exclusively* of loose pyroclastic fragments (pyroclasts, cinders, scoria, lapilli, and volcanic spindle bombs). If a lava flow occurs, it typically breaches the weak base of the cone rather than overflowing the summit crater.
- **Edifice Geometry:** Steep, straight slopes fixed by the internal friction angle (angle of repose $\approx 30^\circ\text{--}35^\circ$) of loose tephra, with a bowl-shaped summit crater. Heights rarely exceed 300–400 meters.
- **Lifespan:** Typically monogenetic (formed during a single eruptive episode lasting months to a few years, e.g., Mexico's Parícutin in 1943–1952).

2.3 Stratovolcanoes (Composite Volcanoes)

Stratovolcanoes are majestic, steep-sided, concave-upward conical mountains located predominantly along subduction-zone volcanic arcs (the "Ring of Fire"):

- **"Composite" Nature:** The term *composite volcano* directly reflects their layered internal architecture: alternating strata of solid lava flows and explosive pyroclastic tephra (ash-fall beds, pumice lapilli, block-and-ash flow deposits, lahars).
- **Magma Chemistry:** Intermediate to felsic calc-alkaline magmas (andesite, dacite, with subordinate basalt and rhyolite) containing moderate-to-high volatile contents ($\text{H}_2\text{O}, \text{CO}_2, \text{SO}_2$).
- **Eruptive Style:** Highly explosive Plinian to Vulcanian eruptions alternating with calmer effusive phases. Slopes reach 30°–36° near the summit and taper to 6°–10° at the apron.

2.4 Lava Domes (Volcanic Domes)

Lava domes (also called coulées or tholoids) are steep-sided, rounded, bulbous knolls:

- **Viscosity Factor:** Formed when highly viscous, gas-depleted, silica-rich lava (dacite, rhyolite, trachyte; $>65\%\text{ SiO}_2$) erupts slowly. The viscosity is so high ($10^8\text{ to }10^{11}\text{ Pa}\cdot\text{s}$) that the magma cannot flow away; instead, it is squeezed out like toothpaste, piling up directly over or within the vent.
- **Structural Hazards:** As the dome inflates internally (endogenous growth) or breaks through its carapace (exogenous growth), steep oversteepened flanks collapse under gravity, triggering devastating block-and-ash pyroclastic flows (e.g., Mount Pelée in 1902; Soufrière Hills, Montserrat).

3. Systematic Comparison of Volcanic edifies

Volcano Type	Primary Lava / Material	Edifice Geometry & Slopes	Eruptive Style	Diagnostic Characteristic (Exam Match)
Shield Volcano	Basalt (low viscosity)	Convex, broad flattened dome (2°–10°)	Effusive Hawaiian flows	**Has shape of a flattened dome (2)**
Cinder Cone	Scoria / cinders (pyroclastic)	Conical hill at angle of repose (30°–35°)	Strombolian gas fountaining	**Formed exclusively of pyroclastic materials (3)**
Stratovolcano	Andesite, dacite, tephra	Concave cone (slope 10°–35°)	Plinian / Vulcanian & flows	**Also known as composite volcano (1)**
Lava Dome	Dacite, rhyolite (very high viscosity)	Steep-sided bulbous mound / plug	Passive extrusion / collapse flows	**Formed by highly viscous lava (4)**

4. Step-by-Step Code Matching

A (Shield volcano) → 2 (Flattened dome shape)
B (Cinder cone volcano) → 3 (Exclusively pyroclastic materials)
C (Stratovolcano) → 1 (Also known as composite volcano)
D (Lava dome volcano) → 4 (Formed by highly viscous lava)

Resulting Pairing: A-2, B-3, C-1, D-4 → Option (c)

Textbook Authority

John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd ed., Chapter 4: Igneous Structures and Field Relationships, pp. 62–64):
"Shield volcanoes: Low viscosity basalt flows produce a landform with a convex upward profile and low slope, producing a flattened dome... Composite volcano, or stratovolcano: Layered sequence of flows and pyroclastics... Pyroclastic cones, such as scoria cones or cinder cones: Composed exclusively of loose fragmental debris... Domes: Formed by the extrusion of highly viscous silicic magmas (dacites and rhyolites)."
P. K. Mukherjee, *Textbook of Geology*:
"Volcanoes are classified as: 1. Shield volcanoes (broad flattened dome of basic lavas); 2. Cinder cones (entirely made of loose cinders and ash); 3. Composite cones or Stratovolcanoes (interstratified lavas and pyroclasts); 4. Volcanic domes (viscous acid lava)."